Task 1: Fine-Grained Bird Classification

In this task, the fine-grained classification of bird species was carried out using two deep learning methods: a transfer learning model implemented using DenseNet121, and a custom CNN model trained from scratch. The dataset was split into 90% for training and 10% for validation, with the test data strictly set aside. Random seeds were set to ensure reproducibility.

For Model 1, DenseNet121 pretrained on ImageNet was used to leverage strong feature representations. In Stage 1, the pretrained backbone was frozen and only the classification head was trained for 20 epochs. This allowed the model to adapt to the bird dataset without modifying pretrained weights. In Stage 2, the entire network was unfrozen and fine-tuned using a smaller learning rate (7e-5) to prevent large weight updates and preserve learned features. The final test accuracy achieved was 66.85%, with weighted precision (0.6807), recall (0.6685), and F1-score (0.6667). The confusion matrix showed strong diagonal dominance, indicating effective class separation.

For Model 2, a tailored CNN architecture with four convolutional layers, Batch Normalization, Global Average Pooling, and Dropout was designed. Data augmentation was used to enhance generalization. The result for this model was a test accuracy of 20.5%. While this was substantially worse than Dense Net, it was better than random guessing, which was around 0.5%. The confusion matrix indicated a higher level of confusion between classes, which indicated poor feature extraction ability relative to the pre-trained model.

In **conclusion**, the experiment proves that transfer learning is an effective technique in improving the performance of fine-grained classification. The performance analysis of both models satisfies the requirements of the coursework.

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